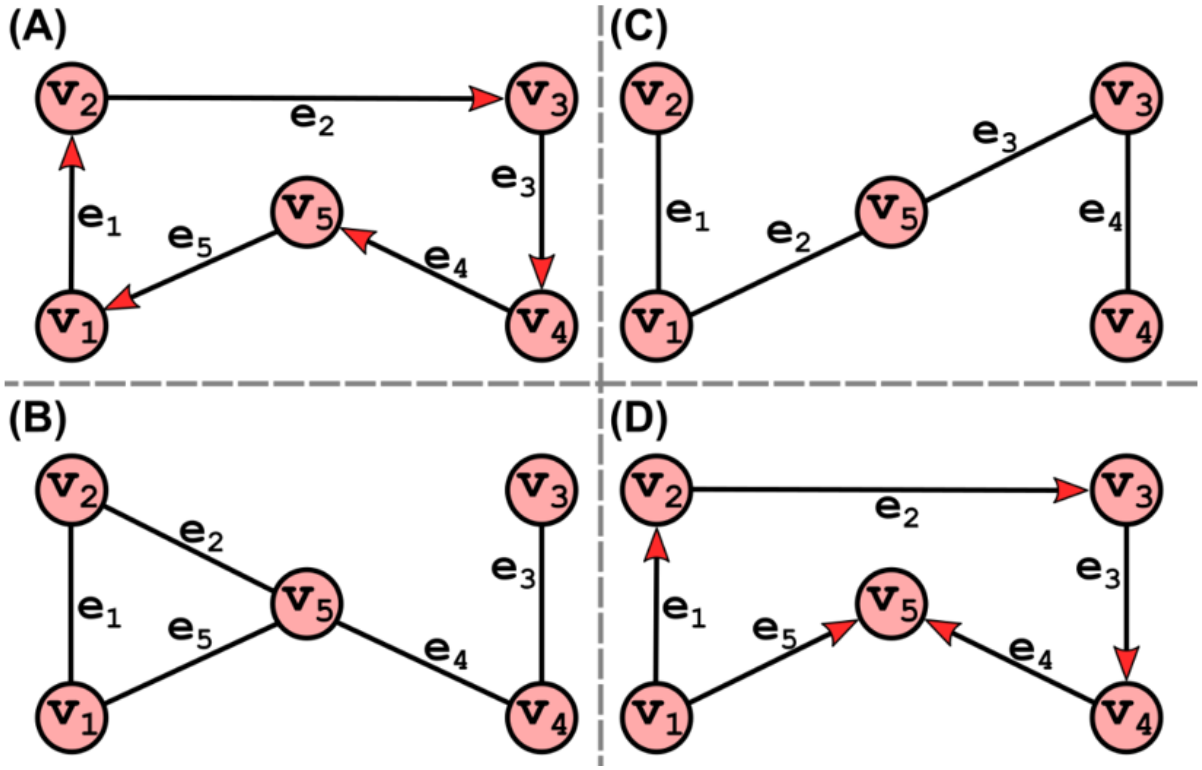


Figure 14.6 Cyclic graphs ((A) directed, (B) undirected) versus acyclic graphs ((C) and (D)).

Given a directed acyclic graph `G=(V,E)`, in fact, a partial ordering is a relation ≤ such that for any couple of vertices `(u,v)`, exactly one of these three conditions will hold:

- u ≤ v, if there is a path of any number of edges starting from u and reaching v;
- v ≤ u, if there is a path of any number of edges starting from u and reaching v;
- u <> v, they are not comparable because there isn't any path from u to v or vice versa.

Figure 14.7 shows a couple of generic DAGs, and a chain: the latter is the only kind of DAG defining a total ordering on its nodes.